MIL-C-49109(EL)

31 December 1976

MILITARY SPECIFICATION
COUNTER, ELECTRONIC, DIGITAL READOUT, TS-3662( )(V)1/U

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers 500 MHz Universal Frequency Counters herein after called equipment.

1.2 Classification. Equipment covered by this specification may be commercially available equipment modified to the extent necessary to meet the requirements as defined in MIL-T-28800, and unless otherwise specified herein, for Type II, Class 5, Style E equipments.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for Bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Command, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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STANDARDS

MILITARY

MIL-STD-781 Reliability Tests Exponential Distribution

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS

IEEE-STD-488-1975 Standard Digital Interface for Programmable Instrumentation

(Application for copies should be addressed to the IEEE Standard Sales, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Frequency Counter. Except for non-operating temperature, equipment covered by this specification shall meet the requirements of this specification and of MIL-T-28800 for Type II, Class 5, Style E unless otherwise specified herein.

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3.1.1 Bid Samples. When required in the solicitation, bid samples shall be submitted to the procuring activity for bid sample evaluation (See 4.3, 6.2).

3.2 Safety. The equipment shall comply with the Type III safety requirements of MIL-T-28800.

3.3 Parts, Materials and processes. Unless otherwise specified herein, the equipment shall comply with the Type III parts, materials and processes requirement as defined in MIL-T-28800.

3.3.1 Restricted Materials. The equipment shall comply with the restricted material requirements of MIL-T-28800 including the requirements covering use of mercury or radio-active materials; and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3.3.2 Signal input and output connectors. Unless otherwise specified herein, all signal input and output connectors shall be Type BNC female.

3.4 Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type III design and construction requirements of MIL-T-28800.

3.4.1 Solid state construction. The equipment shall be of solid state, modular, miniaturized construction. Electron tubes shall not be used.

3.4.2 Controls. Unless otherwise specified herein, built-in adjustments and compensating devices shall not be externally accessible.

3.4.3 Encapsulation and embedment. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits.

3.5 Electrical power sources and connections. The equipment shall comply with the Type III electrical power sources and connections requirements of MIL-T-28800.

3.5.1 Maximum power. The equipment shall not consume more than 150 watts.

3.5.2 Input power switch. An input power switch need not be provided.

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3.5.2.1 Standby-operate switch. A standby switch shall be provided. The "standby" position shall allow only the oven and its associated circuitry to be energized. The operate position shall allow all circuits to be energized.

3.5.3 Power indicator. A front panel indicator shall be provided to indicate when the equipment is energized. Indication shall be provided for both the standby and operate mode.

3.5.4 Fuses and circuit breakers. Fuses or circuit breakers shall be in accordance with the Type III fuse or circuit breaker requirements of MIL-T-28800. The requirement for fusing both sides of each input power line shall not apply. The fuse or circuit breaker shall be provided only in the one conductor normally ungrounded.

3.6 Dimensions, weight and mechanical stability.

3.6.1 Weight. The maximum weight of the equipment shall not exceed 14 Kg (30.87 pounds).

3.6.2 Dimensions. The maximum dimensions of the equipment, excluding a tilt bail handle assembly if provided, shall not exceed 465 mm (18.3 inches) in width, 135 mm (5.31 inches) in height and 465 mm (19.0 inches) in depth. Frontal area shall not exceed 465 centimeters squared (72.2 inches squared).

The tilt bail handle assembly if used shall not increase the width to more than 485 mm (19.09 inches) and the depth to more than 635 mm (25 inches) in the carrying position. Clearance between the front cover and the grip portion of the handle shall be no less than 50 mm (1.97 inches) at its closest proximity.

3.7 Enclosures. The equipment enclosure shall be as specified in MIL-T-28800 for Style E.

3.8 Marking and identification. The equipment shall comply with the Type III marking and identification requirements of MIL-T-28800.

3.9 Environmental requirements. Except for non-operating storage temperature, the equipment shall comply with the Class 5, Style E requirements of MIL-T-28800.

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3.9.1 Non-Operating storage temperature. The equipment shall comply with the Class 5 non-operating storage temperature requirements of MIL-T-28800 except the minimum temperature shall be -40°C.

3.9.2 Electromagnetic interference. (EMI)

3.9.2.1 EMI. The equipment shall comply with the EMI requirements of MIL-T-28800 and as specified in a thru i:

a. CE01: Narrow band requirements from 10 KHz to 20 KHz only.

b. CE03: Full limits.

c. CS01· Full limits.

d. CS02 Full limits.

e. CS06· Full limits.

f. RE01 Relaxed 50 db at the fundamental, third harmonic, and fifth harmonic of the power source frequency.

g. RE02. Relaxed 25 db from 25 MHz to 200 MHz.

h. RS01: Full limits.

i. RS03 Full limits to 1 GHz.

3.9.3 Warm-up time. Warm-up time shall be as specified in 3.12.2.

3.10 Reliability Requirements.

3.10.1 Reliability Burn-In. Each equipment delivered against this specification shall be subjected to a minimum 100 hour on-time burn-in procedure as specified in para. 4.4.6.3. The last 24 hours of burn-in shall be failure free.

3.10.2 Reliability. Reliability shall comply with requirements as specified herein. The specified MTBF shall be 5,000 hours when tested as specified in para 4.4.6. A failure shall be as defined in MIL-STD-781, para 3.1, and as any departure from the

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required performance or operation outside of the required accuracies (not correctable by normal use of the operating controls) after the test is initiated. Test level B, MIL-STD-781, shall be the required test level.

3.11 Maintainability requirements. The equipment shall comply with the Type III maintainability requirements of MIL-T-28800 and as specified herein (See 6.2.3.2).

3.11.1 Certification time. Certification time shall not exceed 2.0 hours, and it shall include all time required to check all inputs and outputs, on all range and function settings.

3.11.2 Maintenance provisions The equipment shall meet the Type III maintenance provisions of MIL-T-28800 except the accessibility of accessories does not apply.

3.11.2.1 Maintenance and calibration aids. When required in the contract or order, any circuit board extenders, special adaptors, special tools, and patch cables required for maintenance or calibration of equipment shall be stowed in a suitable reusable pouch or container which shall be marked "MAINTENANCE AIDS FOR USE WITH (equipment nomenclature)". (See 6.2.1.1)

3.11.3 Indicator lights. Unless approved by the procuring activity upon presentation of acceptable reliability data, indicator lights other than light emitting diodes (LEDS) shall be accessible from the operator's side of the front panel.

3.11.4 Washability. The equipment shall comply with the washability requirements of MIL-T-28800.

3.12 Performance requirements. The equipment shall contain the following electrical performance requirements

a. Environmental performance (3.12.1)

b. Warm up performance (3.12.2)

c. Time base (3.12.3)

d. Display (3.12.4)

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e. Input characteristics	(3.12.5)
f. Frequency Measurement	(3.12.6)
g. Time Interval Measurement	(3.12.7)
h. Period Measurement	(3.12.8)
i. Ratio Measurement	(3.12.9)
j. Outputs	(3.12.10)
k. Logic interface	(3.12.11)

3.12.1 Environmental performance. Equipments shall meet the requirements specified herein under all combinations of environments specified herein.

3.12.2 Warm up performance. Warm up time shall not exceed one minute after the operate mode is selected when sufficient time has previously elapsed in the standby mode to permit the time base to stabilize to the degree required for the measurement in accordance with 3.12.3.

3.12.3 Time base. Time base specifications are given in terms of fractional frequency error (ffe) (See 6.3.1). Except for temperature stability all time base requirements are specified for 25°C. Time base requirements shall be measured from the 10 MHz standard output with the equipment in the check mode and the equipment calibrated in accordance with 3.12.3.7. Time base requirements are also based on 72 hours of non-operation preceded by 72 hours of operation except as noted.

3.12.3.1 Short term stability. The short term stability of the oscillator shall be less than 6×10^{-11} root mean square (rms) for 40 adjacent samples. Short term stability shall be measured after power has been applied to the oscillator and associated circuitry for a period of 5 hours of operation preceded by 72 hours of operation preceded by 72 hours with no source applied.

3.12.3.2 Long term stability (aging rate). The oscillator aging rate shall be less than $\pm 2 \times 10^{-9}$/day after 48 hours of operation and $\pm 1 \times 10^{-9}$/day after 96 hours of operation.

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3.12.3.3 Frequency retrace. The oscillator ffe 30 minutes after application of source power following 24 hours of non-operation shall be within $\pm 2 \times 10^{-8}$. For one hour, four hours and 24 hours of non-operation, the ffe shall be within $\pm 1.5 \times 10^{-8}$, $\pm 6 \times 10^{-9}$ and 1.5×10^{-9} respectively.

3.12.3.4 Frequency restabilization (aging recovery).

3.12.3.4.1 Short term settling. The value of short term stability 30 minutes after application of source power shall be less than 1×10^{-9} rms for 40 adjacent one second samples.

3.12.3.4.2 Restabilization. The frequency measured at five hours after application of source power shall be within $\pm 1 \times 10^{-8}$ of the frequency at 48 hours after application of source power.

3.12.3.5 Stability during power change. The oscillator ffe shall be no greater than 2×10^{-10} within two minutes after any power change within the specified steady state band.

3.12.3.6 Stability during temperature change. The oscillator ffe shall be within 1.4×10^{-8} for any incremental or total temperature change within the specified operating range.

3.12.3.7 Oscillator adjustments. The oscillator shall be capable of being adjusted to within one part in 10^9 of an external standard as measured at the 10 MHz standard output connector. The total range of adjustment shall be at least one part in 10^6.

3.12.3.8 Total ffe. The total ffe of the oscillator at the end of a twelve month period for all time base specifications shall not exceed $\pm 5.0 \times 10^{-7}$.

3.12.3.9 External standard input. The equipment shall be capable of specified operation from any 10 MHz external frequency standard having a value of one volt rms into 50 ohms. When the external standard is used it shall be in lieu of the internal time base oscillator.

3.12.4 Display. All display devices shall be visible light or glow devices readable at a distance of 15 feet through a subtended angle of 45 degrees in an ambient light

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level of 100 foot candles. Each display device shall have a minimum life expectancy of at least 50,000 operating hours. Provisions shall be incorporated to test all light emitting segments of the display devices.

3.12.4.1 Numeric readout. The numeric readout shall be a 9 character, in-line, horizontal, single plane, digital display at least 10 mm (0.4 inches) high.

3.12.4.1.1 Decimal point. The correct decimal point shall be a display device and shall be automatically displayed.

3.12.4.1.2 Measurement units. The correct measurement unit display device in accordance with a through f shall be presented in conjunction with the numeric display.

a. MHz

b. KHz

c. Hz

d. usec

e. msec

f. sec

3.12.4.1.3 Display storage. The numeric readout shall be continuous and change only when the measurement is different from that counted during the previous measurement cycle.

3.12.4.1.4 Display blanking. All non-significant leading zeros shall be blanked. All readout digits below the selected resolution of the displayed frequency shall not be displayed.

3.12.4.2 Gate lamp. A front panel display device shall indicate gate opening and closing.

3.12.4.2.1 Reset. The reset control shall allow manual initiation of the gate.

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3.12.4.3 Sample rate. The sample rate shall be variable from 0.01 to at least 5 seconds. The sample shall be capable of being held indefinitely. Erroneous data resulting from conflicts between sample rate and gate time shall not be displayed.

3.12.4.4 Self check. The self check function shall be provided to verify functional operation of the frequency measurement, time interval measurement, period measurement, and ratio measurement modes in all resolution positions.

3.12.5 Input characteristics. Three input channels shall be provided. Two channels shall have approximately identical characteristics and shall be provided for high impedance use. The third channel shall be provided for 50 ohm use. For clarity only, the high impedance channels are identified as channel A and channel B and the 50 ohm channel is identified as channel C.

3.12.5.1 Common input. Provision shall be made for selection of common or separate input. During common input, the signal applied to a single channel shall be applied to both channels A and B simultaneously, isolating the alternate (A or B) channel.

3.12.5.2 Attenuators. Separate attenuators providing attenuation of x1, x10 and x100 shall be provided for channels A and B.

3.12.5.3 Sensitivity. The sensitivities for channels A and B shall be at least 50 mv from dc to 100 MHz for sine waves and at least 140 mv peak to peak for pulses with a duty cycle of 50 percent. The sensitivity for channel C shall be at least 15 mv rms from 50 MHz to 500 MHz for sine waves and at least 50 mv peak to peak for pulses with a duty cycle of 50 percent.

3.12.5.4 Dynamic range. The dynamic ranges of channels A and B shall be at least 50 mv rms to 500 mv rms times the attenuator setting from dc to 100 MHz. The dynamic range of channel C shall be at least 15 mv rms to 500 mv rms from 50 MHz to 500 MHz.

3.12.5.5 Maximum input. Maximum input without damage shall be in accordance with Table I.

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3.12.5.6 Protection. The 50 ohm input shall be protected to prevent damage from input signals with levels up to at least 200 V peak. Protective devices designed to open circuit permanently to comply with this requirement shall be replaceable from the front panel without the need for special tools.

3.12.5.7 Input impedances. The input impedances of channels A and B shall each be at least 1 megohm shunted by not more than 35 picofarad (pF). When channels A and B are shunted in common input configuration (See 3.12.5.1), the input impedance shall be at least 500 K ohms shunted by not more than 70pF. The input impedance of channel C shall be 50 ohms nominal.

3.12.5.8 Coupling. Channels A and B shall be provided with ac and dc coupling. Channel C shall be provided with ac coupling.

3.12.5.9 Triggering.

3.12.5.9.1 Trigger slope. Channels A and B shall be provided with independent selection of triggering for both the positive or negative slope of the input signal.

3.12.5.9.2 Trigger level. Channels A and B shall be provided with independent selection of any trigger level at any point from plus 2.5 volts times the attenuator setting through zero to minus 2.5 volts times the attenuator setting. Channels A and B shall also be provided with a preset position for triggering at zero volts. Triggering of channel C shall be preset at zero volts.

3.12.5.9.3 Trigger level stability. Channel A and B trigger level with a 1 KHz signal applied and channel C trigger level with a 50 MHz signal applied shall not drift more than 25 mV during any two hour period at any constant temperature and an additional 25 mv wher the temperature is repeatedly cycled over the temperature operating range.

3.12.5.10 Arming. Provisions shall be made to arm the equipment for any required measurement at the option of the operator. When arming is activated, the measurement channel (either A, B or C) is enabled by another channel. The arming signal shall initiate the measurement and the length of the measurement shall be controlled by the time base setting.

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TABLE I - MAXIMUM INPUT WITHOUT DAMAGE

CHANNEL	ATTENUATOR SETTING	FREQUENCY RANGE	AC COUPLED	DC COUPLED
A and B	X1	0 to 20 Hz	200V (DC + Peak AC)	250V rms
		20 Hz to 50 KHz	250V rms	250V rms
		50 KHz to 2.5 MHz	$\frac{1.25 \times 10^7}{\text{frequency}}$ V rms	$\frac{1.25 \times 10^7}{\text{frequency}}$ V rms
		2.5 MHz to 100 MHz	5V rms	5V rms
	X10	0 to 20 Hz	200V (DC + Peak AC)	250V rms
		20 Hz to 500 KHz	250V rms	250V rms
		500 KHz to 2.5 MHz	$\frac{1.25 \times 10^8}{\text{frequency}}$ V rms	$\frac{1.25 \times 10^8}{\text{frequency}}$ V rms
		2.5 MHz to 100 MHz	50V rms	50V rms
	X100	0 to 100 MHz	250V rms	250V rms
C	X1.	50 MHz to 500 MHz	5V rms	N/A

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3.12.6 Frequency measurements. The equipment shall automatically measure and display the frequency of electrical signals by the direct count technique.

3.12.6.1 Frequency range. The frequency ranges of channels A and B shall be at least 0 to 100 MHz for DC coupling and at least 20 Hz to 100 MHz for AC coupling. The frequency range of channel C shall be at least 50 MHz to 500 MHz. The equipment shall accept and properly measure sine wave and pulse signals that have a pulse width of at least 5 nsec for channel A and B and least 1 nsec for channel C.

3.12.6.1.1 Burst. The equipment shall be capable of automatically measuring the freauency of the r.f. signals composing any burst when the r.f. signals are within the ranges of 3.12.6.1 and the burst length is at least 2 usec long for channels A and B and 8 usec for channel C.

3.12.6.2 Frequency Accuracy. The accuracy of frequency measurement shall be within ± 1 count ± time base error.

3.12.6.3 Frequency Resolution. The equipment shall have a selectable resolution from 0.1 Hz to 1 MHz in decade steps. The maximum time allowed for a 1 Hz change in the input signal to be displayed when the resolution is 1 Hz, is 1 second.

3.12.7 Time interval measurements. The equipment shall automatically measure and display the time interval between events of electrical signals.

3.12.7.1 Time internal range. The time internal range shall be from at least 0.1 usec to 10^6 seconds.

3.12.7.1.1 Minimum pulse width. The time interval start and stop shall be capable of being triggered with pulses of 25 nsec or greater. False non-zero values shall not be displayed.

3.12.7.2 Time interval accuracy. The accuracy of time interval measurement shall be within ± 1 count ± time base error ± trigger error (See 6.3.3).

3.12.7.3 Time interval resolution. The equipment shall have a selectable resolution from 100 nsec to 1 sec in decade steps for single time interval measurements.

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3.12.8 Period measurements. The equipment shall automatically measure and display the period and the period average of electrical signals.

3.12.8.1 Period range. The frequency range of input signals for period and period average measurements shall be at least from zero to 10 MHz.

3.12.8.2 Period accuracy. The accuracy of period measurement shall be within ±1 count ± time base error ± (trigger error divided by number of periods averaged) (See 6.3.3).

3.12.8.3 Period resolution. The equipment shall have a selectable resolution from 100 nsec to 1 sec in decade steps for single period and 1 psec to 100 nsec in decade steps for period average.

3.12.9 Ratio measurements. The equipment shall automatically measure and display the ratio and the ratio average of two signals herein identified as f_1/f_2 where f_1 is the signal applied to the decimal counters through channels B and C and f_2 is the freauency used as a gating signal through channel A (See 3.12.5).

3.12.9.1 Ratio range. The frequency range shall be at least from zero to 500 MHz for f_1 and zero to 10 MHz for f_2 when using channels C and A, zero to 100 MHz for f_1 and zero to 10 MHz for f_2 when using channels B and A.

3.12.9.1.1 Ratios averaged. The ratios averaged shall be selectable from 1 to 10^7 in decade multiples of f_2.

3.12.9.2 Ratio accuracy. The accuracy of the ratio measurement for 2 or more events of f_2 shall be within ± 1 count of f_1, ± the trigger error of f_2 times the frequency of f_1.

3.12.10 Outputs.

3.12.10.1 Standard frequency output. The equipment shall provide a 10 MHz standard frequency output which is independent of the time base switch setting and uninterrupted during all functions. The output voltage shall be at least one volt rms into an external impedance of 50 ohms.

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3.12.10.2 Gate output. The equipment shall provide a gate output during and only during the frequency measurement.

3.12.10.3 Marker output. The equipment shall provide a marker output during and only during the time interval measurement period.

3.12.11 Logic interface. A logic interface shall be provided in accordance with IEEE 488-1975.

3.12.11.1 Function programming characteristics. All measurement data, including multipliers and units, shall be transferred by means of the logic interface. All control and switch functions except for the following shall be programmable.

a. Source power selection.

b. Frequency standard selection.

c. Standby/operate selection.

3.12.11.1.1 Trigger level programming. Trigger level programming shall be in 10 millivolt increments.

3.12.11.1.1.1 Trigger point settability. The trigger point shall be within ±35 millivolts of the programmed trigger level.

3.12.11.1.2 Sample programming. Under interface control the equipment shall have two sampling modes (as selected by the controller). In the first mode the equipment shall not make any measurement until commanded to do so by the controller. The rate at which the controller can command the equipment to make measurements shall be at least 100 measurements per second or the maximum rate possible with the required resolution.

In the second mode the equipment shall continually make measurements on its own at a rate of at least 100 measurements per second or at the maximum rate possible with the required resolution.

3.12.11.2 Interface characteristics. The equipment shall include the interface functions as in a thru f.

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a. Source handshake interface function (SH), subset identification - SH1.

b. Acceptor handshake interface function (AH), subset identification - AH1.

c. Talker interface function (T), subset identification - T1.

d. Listener interface function (L), subset identification - L2.

e. Service request interface function (SR), subset identification - SR1.

f. Remote local interface function (RL), subset identification - RL1.

3.12.11.2.1 Numeric data. Unless otherwise specified in the contract or order, all numeric data (input and output) shall be ASCII and the most significant digit shall be transferred first.

3.12.11.2.2 Alpha-numeric data. Unless otherwise specified in the contract or order, all alpha-numeric data (input and output) shall be ASCII.

3.12.11.2.3 Address assignment. Complete five bit switchable address capabilities shall be provided for the talker function (T1) address (MTA) and the listener function (L2) address (MLA) on the front panel for Configuration A and on the rear panel for Configurations B, C, and D. If a single device for address switching is provided for both the MLA and the MTA, then the five bit switchable address for the MLA shall be equal to that for the MTA.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Bid sample testing will be performed by the Government.

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4.1.1 Quality assurance terms and definitions. Quality assurance terms used herein shall be in accordance with MIL-T-28800.

4.2 Classification of inspections. The inspections required herein are classified as in a through b.

a. Bid Sample inspection (See 4.3).

b. Quality conformance inspection (See 4.4).

4.2.1 Bid sample inspection groups. Bid sample examinations and tests are classified into groups in accordance with Table II, 4.3, and a through f.

a. Bid sample inspection, Group A (See 4.3.1).

b. Bid sample tests, Group B (See 4.3.2).

c. Bid sample tests, Group C (See 4.3.3).

d. Bid sample tests, Group D (See 4.3.4).

e. Bid sample tests, Group E (See 4.3.5).

f. Bid sample tests, Group F (See 4.3.6).

4.2.2 Quality conformance inspection groups. Quality conformance examinations and tests are classified into groups in accordance with Table II, 4.4, and a through f.

a. Production inspection, Group A (See 4.4.1).

b. Production control tests, Group B (See 4.4.2).

c. Environmental and electrical power tests, Group C (See 4.4.3).

d. EMI tests, Group D (See 4.4.4).

e. Other tests, Group E (See 4.4.5).

f. Reliability tests, Group F (See 4.4.6).

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4.3 Bid sample inspection. When required in the solicitation, bid samples shall be submitted to the procuring activity for bid sample evaluation. Unless otherwise required in the solicitation the bid sample shall consist of two equipments, including accessories, instruction manuals and clear and concise rationale showing how the reliability and maintainability characteristics of the equipment comply with the requirements of the solicitation (See 6.4). In the event an equipment under test becomes unacceptable during the Group C or Group D testing, the other equipment will be subjected to the Group C or Group D tests which were not satisfactorily completed.

4.3.1 Bid sample inspection, Group A. The two equipments will be subjected to the Group A inspection specified in Tables II and III.

4.3.2 Bid sample tests, Group B. The two equipments that have satisfactorily passed the Group A inspection will be subjected to the Group B tests specified in Tables II and IV.

4.3.3 Bid sample tests, Group C. One equipment that has satisfactorily passed the Group B tests will be subjected to the Group C tests as specified in Table II.

4.3.4 Bid sample tests, Group D. One equipment that has satisfactorily passed the Group B tests will be subjected to the Group D tests as specified in Table II.

4.3.5 Bid sample tests, Group E. Equipments that have satisfactorily passed the Group B tests will be subjected to the Group E tests as specified in Table II. The number of equipments to be tested will be as in a through c.

a. Maintainability. No equipment will be tested.

b. Acoustical noise test. One equipment will be tested if forced air cooling is used.

c. Dimensions and weight tests. One equipment will be tested.

4.3.6 Bid sample tests, Group F. Unless otherwise specified in the solicitation, Group F reliability testing will not be performed on bid samples.

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TABLE II - EXAMINATION AND TEST GROUPS

Description	Requirement	Inspection Method	
		Bid Sample	Quality Conformance
Group A			
Preoperational inspection	3.	4.5	4.5
Level A performance	3.12	4.5	4.5
Leakage current	3.2	4.5	4.5
Group B			
Level B performance	3.12	4.5	4.5
Group C			
Temperature and humidity			
Humidity	3.9	4.5	4.5
Low temperature	3.9	4.5	4.5
High temperature	3.9	4.5	4.5
Temperature stability time base	3.12.3.6	4.5	4.5
Altitude	3.9	4.5	4.5
Vibration	3.9	4.5	4.5
Shock			
Shock pulse	3.9	4.5	4.5
Bench handling	3.9	4.5	4.5
Electrical power			
Input power consumption	3.5	4.5	4.5
Voltage and frequency variation	3.5	4.5	4.5
Voltage transient	3.5	4.5	4.5
Frequency transient	3.5	4.5	4.5
Line voltage stability time base	3.12.3.5	4.5	4.5
Verification, Group C	3.12	4.5	4.5
Group D			
Electromagnetic interference	3.9.2	4.5	4.5

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TABLE II - EXAMINATION AND TEST GROUPS (Contd)

Description	Requirement	Inspection Method Bid Sample	Inspection Method Quality Conformance
Group E			
Maintainability	3.11	4.5	4.5
Washability	3.11.4	4.5	4.5
Acoustic noise	3.9	4.5	4.5
Dimensions	3.6	4.5	4.5
Weight	3.6	4.5	4.5
Group F			
Reliability	3.10	4.5	4.5

4.4 Quality conformance inspection. Unless otherwise required herein the test methods, procedures, and conditions used for quality conformance inspection shall be as specified in 4.5 and MIL-T-28800.

4.4.1 Production inspection. Each equipment shall be subjected to the Group A inspection as specified in Tables II and III to ensure qualitatively the proper functioning of the equipment, including all operating controls, and conformance with safety requirements.

4.4.2 Production control tests. The Group B tests of Tables II and IV shall be performed on equipment that have passed Group A inspection.

4.4.2.1 Production control sampling plan. From each lot which has passed production inspection (See 4.4.1) a random sample shall be selected and subjected to the Group B tests. Minimum sampling shall be in accordance with MIL-T-28800.

4.4.3 Environmental and electrical power tests. The Group C tests of Table II shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

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4.4.3.1 Environmental and electrical power sampling plan. From the First Production Lot, which has passed the required tests in accordance with 4.4.2 a random sample shall be selected and subjected to the Group C tests. The sampling shall be in accordance with MIL-T-28800.

4.4.4 EMI tests. The Group D tests of Table II shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.4.1 EMI sampling plan. From the First Production lot, which has passed the required tests in accordance with 4.4.2, a random sample shall be selected and subjected to the Group D tests. The sampling shall be in accordance with MIL-T-28800.

4.4.5 Other tests. The Group E tests of Table II shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.5.1 Other tests sampling plan. From the First Production lot, which has passed the required tests in accordance with 4.4.2, a random sample shall be selected and subjected to the Group E tests. The sampling shall be in accordance with a through c.

a. Acoustic noise. One equipment shall be randomly selected from the first production lot and tested if forced air cooling is used.

b. Maintainability. One equipment shall be randomly selected from the first production lot and tested.

c. Dimensions and weight. Two equipment shall be randomly selected from the first production lot and tested.

4.4.6 Reliability tests. The Group F tests shall be conductec in accoraance with the following test plan

θ_0 = 5000 θ_1 = 2500 $\alpha = \beta$ = .20

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TOTAL TEST TIME*

Failures	Accept ** (equal or more)	Reject *** (equal or less)
0	4000	N/A
1	7500	1115
2	10750	4125
3	13750	7700
4	16750	11450
5	19750	15450
6	22750	22750 1

*Total test time is total unit hours of equipment on time and is expressed in multiples of specified MTBF.

**Accept if test time is greater than or equal to that listed.

***Reject if test time is less than or equal to that listed.

1 Reject if test time is strictly less than 22750 hrs. with 6 failures.

4.4.6.1 Reliability Demonstration Phase. The reliability demonstration shall be performed on a random sample of ten equipments selected from the first production lot.

4.4.6.2 Reliability sampling phase. Unless the reliability has been demonstrated in accordance with paragraph 4.4.6.1 during Bid Sample, the demonstration should be run on only the first production lot.

4.4.6.2.1 Duty Cycle. The equipment shall be operated weekly in the satisfactory operation test, paragraph 4.5b. The equipment shall be performing continuously in the satisfactory operation check, paragraph 4.5a, during the power-on time.

4.4.6.2.2 Reliability Examination and Test Method. Examination and Test methods, of the specification shall be used to verify equipment operation on a continuous basis. Satisfactory operation check, paragraph 4.5, subparagraph a, shall be used to check a one channel and the display section Satisfactory operation test shall be used to confirm specified operation prior to reliability demonstration and at completion. Weekly satisfactory operation test, paragraph 4.5b, specified shall confirm all

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parameters found in Level A, paragraph 3.12, Performance Requirements, and Table III of this specification.

4.4.6.3 Reliability Burn-In. Each deliverable equipment shall be subjected to a minimum 100 hour on-time burn-in period prior to Group A testing as follows

4.4.6.3.1 Each equipment will be allowed to accumulate failures during the first 76 hours of burn-in time.

4.4.6.3.2 During the last 24 hours of burn-in, each equipment must operate failure free. Each equipment which fails during this period shall be repaired and returned to test until it successfully survives a 24 hour period without failure.

4.4.6.3.3 Failures which occur during the burn-in test shall be noted and reported, but shall not be counted toward the establishment of equipment MTBF.

4.4.6.3.4 Duty cycle imposed and parameters shall be measured during the 100 hour on time burn-in. Prior to the burn-in a satisfactory operation test shall be conducted. Daily satisfactory operation checks shall be conducted. For the last 24 hour failure free period, a complete satisfactory operation test shall be conducted after the period. Satisfactory operation test is as defined in paragraph 4.5 of this specification.

4.4.6.3.5 Test environment utilized during burn-in shall be the same as the reliability demonstration environment, Test Level B, of MIL-STD-781. In addition, input voltage cycling test in accordance with MIL-STD-781 shall be included.

4.4.6.4 Reliability failure actions, Group F. In the event of failure(s) during reliability testing, failure action shall be taken as required in MIL-STD-781, paragraph 5.5.

4.4.6.5 Corrective Action. In the event the reliability test reaches a reject decision, corrective action shall be taken as required in MIL-STD-781, paragraph 5.7.

4.5 Examination and test methods. Examination and test methods shall be in accordance with MIL-T-28800 and as specified herein. (See Tables II, III and IV). Except for non-operating storage range, the equipment shall be tested for the Class 5 requirements of MIL-T-28800 (See 3.9.1). Equipment operation shall be verified in accordance with a and b below:

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a. Satisfactory operation check. The satisfactory operation check shall confirm the proper operation of the internal time base, the input circuits of at least one channel, and the display section of the equipment.

b. Satisfactory operation test. The satisfactory operation test shall confirm the specified operation of the equipment for all parameters of the Level A performance test (See Table III).

4.6 Optional quality assurance system. The procuring activity may substitute a Department of Defense (DoD) approved supplier's quality assurance system for any or all tests in Table II.

TABLE III - LEVEL A PERFORMANCE TESTS

DESCRIPTION	Requirement
Display section	3.12.4
Input characteristics	3.12.5
Frequency measurements	3.12.6
Time internal measurements	3.12.7
Period measurements	3.12.8
Ratio measurements	3.12.9
Standard frequency output	3.12.10.1
Gate output	3.12.10.2
Marker output	3.12.10.3
Logic interface	3.12.11

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TABLE IV - LEVEL B PERFORMANCE PARAMETERS

PARAMETER	REQUIREMENT
Warm Up	3.12.2
Time Base	3.12.3
Short Term Stability	3.12.3.1
Long Term Stability	3.12.3.2
Fast Warm Up	3.12.3.3
Frequency Retrace	3.12.3.4
Oscillator Adjustment	3.12.3.7
Adjustment Settling Time	3.12.3.7.1
External Standard Input	3.12.3.9
Display Section	3.12.4
Readout	3.12.4.1
Display Blanking	3.12.4.2
Decimal Point	3.12.4.3
Measurement Units	3.12.4.4
Display Storage	3.12.4.5
Sample Rate	3.12.4.6
Resolution	3.12.4.7
Gate Lamp	3.12.4.8
Reset	3.12.4.9
Self Check	3.12.4.10
Input Characteristics	3.12.5
Common Input	3.12.5.1
Attenuators	3.12.5.2
Sensitivity	3.12.5.3
Dynamic Range	3.12.5.4
Maximum Input	3.12.5.5
Protection	3.12.5.6
Input Impedance	3.12.5.7
Coupling	3.12.5.8
Trigger Slope	3.12.5.9.1
Trigger Level	3.12.5.9.2
Trigger Level Stability	3.12.5.9.3
Arming	3.12.5.10

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TABLE IV - LEVEL B PERFORMANCE PARAMETERS (Contd)

PARAMETER	REQUIREMENT
Frequency Measurements	3.12.6
Frequency Range	3.12.6.1
Burst	3.12.6.1.1
Frequency Accuracy	3.12.6.2
Frequency Resolution	3.12.6.3
Time Interval Measurements	3.12.7
Time Interval Range	3.12.7.1
Minimum Pulse Width	3.12.7.1.1
Time Interval Accuracy	3.12.7.2
Time Interval Resolution	3.12.7.3
Period Measurements	3.12.8
Period Range	3.12.8.1
Period Averaged	3.12.8.1.1
Period Accuracy	3.12.8.2
Period Resolution	3.12.8.3
Ratio Measurements	3.12.9
Ratio Range	3.12.9.1
Ratio Accuracy	3.12.9.2
Outputs	3.12.10
Standard frequency output	3.12.10.2
Gate output	3.12.10.4
Marker output	3.12.10.6
Logic interface	3.12.11

5. PACKAGING

5.1 Preservation, packaging packing, and marking. Preparation for delivery shall be in accordance with MIL-T-28800 and as specified by the procuring activity.

6. NOTES

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6.1 Intended use. Equipment described by this specification is intended for use in testing equipment and systems in laboratory, shop, repair, and depot activities. This specification will be used to procure quality, commercial test equipment modified for, or possessing, specific military requirements.

6.2 Ordering data. The procurement documents should specify:

6.2.1 Title, number, and date of this specification.

6.2.1.1 Maintenance and calibration aids if required.

6.2.1.2 Bid samples.

6.2.1.3 Levels of packing and packaging.

6.2.1.4 Warranty.

6.2.2 Contract data requirements.

6.2.2.1 Nomenclature assignment.

6.2.2.2 National stock number.

6.2.2.3 Bid sample test plans.

6.2.2.4 Test equipment approval requests.

6.2.2.5 Pretest performance records.

6.2.2.6 Maintainability test records.

6.2.2.7 Procedure for water-washing equipment.

6.2.2.8 Reliability test records.

6.2.2.9 Quality assurance test plans.

6.2.2.10 Quality assurance test reports.

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6.2.2.11 Fungus inert materials certification (See 3.9).

6.2.3 Acquisition management systems.

6.2.3.1 Reliability (See 3.10).

6.2.3.2 Maintainability (See 3.11).

6.2.3.3 Calibration system (See 4.5).

6.3 Definitions.

6.3.1 Fractional frequency error. Fractional frequency error is defined as the change (produced by any cause) in freauency divided by the average frequency (before the cause). For example in determing the line voltage stability, if the standard output frequency were 1×10^7 MHz and if the standard output frequency varied by 10 Hz for any line voltage change within the specified steady state band then the ffe for the line voltage would be 10 divided by 1×10^7 which is equal to 1×10^{-6}.

6.3.2 Short term stability. The short term stability is defined as the Allan variance of the oscillator under test divided by the average frequency of the oscillator under test.

f_o = average frequency of oscillator under test

$\overline{f}_k$ = k^{th} frequency sample (a frequency sample is a sample of the frequency averaged over 1 second)

M = quantity of adjacent frequency samples. (M = 60 for this specification)

$$\text{short term stability} = \frac{1}{\overline{f}_o}$$

6.3.3 Trigger error. The maximum allowable trigger error is defined as not greater than 2 times the peak noise voltage divided by the signal slope. Thus, for a signal having a 40 dB signal to noise ratio, the trigger error should not exceed 0.0025 usec divided by the signal slope. The signal slope is expressed as the numerical value (without units) of the voltage per microsecond.

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6.4 Bid sample rationale.

6.4.1 Reliability rationale. Reliability rationale submitted with the bid sample should provide clear and concise rationale showing how the reliability of the equipment complies with minimum requirements of the solicitation. The rationale must include quantitative data. The sources of the data may be from previous reliability tests, reliability predictions or any other quantitative data available to the offeror. Reliability data should include conditions under which data was generated and the failure definitions used. This may include data from previous reliability tests, reliability predictions and other data available to the offeror. In the absence of such data on the equipment being offered, such data on similar equipment of equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and accomplished with specifications and technical literature on the similar equipment. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production reliability testing.

6.4.2 Maintainability rationale. Maintainability rationale submitted with the bid sample should provide clear and concise rationale showing how the maintainability of the equipment complies with requirements of the solicitation. This may include data from previous maintainability tests, records of repair and calibration data, and other data on the equipment available to the offeror. Data on selection of module size and other design (See 3.4) features relevant to maintainability may also be submitted. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production maintainability testing.

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INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

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ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes" in what way?)

4 REMARKS (Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

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